

# To-Do List

May 4, 2026

Three Research Questions:

1. How useful is machine learning modelling in clinical prediction in the Emergency Department with respect to the metrics:
  - Focus on the accuracy of the models;
2. In the Emergency Department, which features are important for the accurate prediction and why?
3. Do prediction models show demographic bias, and can it be corrected?.

## Tables and Figures

- figures 1-3 are not for the article. Remove them
- Part of figure 4 and figure 5 can be kept, but clearly prepared.
- **Table 1:** Features - lists all 29 admission-available features with descriptions and derivation methods
- **Table 2:** Model performance - full comparison of 12 models across ED-LOS, admission, and Hosp-LOS tasks. You can decide to use only 4 models. Three tree-based models and a voting ensemble. You can make the model performance tables to be in 2 pages with long tables or even slanted tables.
- **Figure 1:** SHAP summary - three-panel figure showing top features for XGBoost (ED-LOS), LightGBM (admission), and Random Forest (Hosp-LOS)
- **Tables 3-5:** Demographic bias analysis - one table per task using the best model for that task (LightGBM/ED-LOS, XGBoost/admission, Random Forest/Hosp-LOS)
- **Table titles:** Must include (a) model name, (b) prediction task, (c) metric reported

## Revisions and Action Items

- All claims in the abstract should be consistent with the main body. Please target the abstract to be less than 200 words.
- Include the workflow diagram, but redesign it to show the essence of the work.

- In your model results table, your F1 score does not correspond. Check if there is a bug in the code. Debug...
- Remove feature ablation because the reviewer can ask why you didn't remove the harmful features if you classify them. The SHAP plots are enough to see the feature contributions.
- Remove the best model table. You can highlight them in the main model table.
- Keep the table on ESI.
- Present the CI in the model and metrics directly.
- Which model is the baseline for the model performance by race?
- How did you adjust the threshold for bias?
  - Run student t-test & others like p-value adjustment.
- Maintain the table on the best features but it will be wise to clearly state for which model. Then you can show the consensus features for all the models.
- Two tasks involve multi-class classification. But you reported on the binary. Be explicit in the report so that it will be synchronous. Those are the major points the reviewer will check.
- Remove Table 8 & 9.